

Project Title: Donor and Donation Data Analysis using SQL

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Executive Summary

This report presents a comprehensive SQL-based analysis of donor and donation data. The project was designed to support fundraising decision-making through the identification of donor trends, donation behaviors, demographic patterns, and opportunities for improved engagement. The integrated dataset contains 1000 donor records. Total donations reached \$249,085 with an average contribution of \$249.09.

The analysis evaluated donor demographics, geographic distribution, donation frequency, and data quality. SQL queries were used to join datasets, perform aggregation, calculate descriptive statistics, identify high-value donor segments, and generate business insights. Results indicate meaningful variation in donation behavior across donor groups and regions. Recommendations focus on donor retention, segmentation, data governance, and targeted fundraising strategies.

Business Context & Problem Statement

Non-profit and fundraising organizations depend heavily on donor contributions. Understanding who donates, how frequently they donate, and which donor groups generate the highest value is essential for improving fundraising effectiveness.

The business objective of this project is to transform raw donor and donation records into actionable insights that can support fundraising strategy. Key business questions include:

- Which donor segments contribute the most revenue?
- Are there geographic differences in donation behavior?
- How does donation frequency affect total fundraising performance?
- What data quality issues may affect decision-making?
- What actions can improve donor retention and future donation outcomes?

SQL was selected because it provides an efficient and scalable method for extracting, cleaning, transforming, and analyzing structured donor data.

Data Overview & Schema

Two datasets were used in this project:

1. Donation_Data

- Contains donor demographic information and donation amounts.
- Includes variables such as gender, state, job field, and donation value.

2. Donor_Data

- Contains donor profile characteristics and behavioral information.
- Includes donation frequency, university attended, preferred vehicle, favorite color, movie genre, and second language.

The tables are linked using a common ID field, creating a one-to-one relationship. This relationship enables integrated donor analysis by combining demographic, behavioral, and financial attributes.

Data Dictionary & Assumptions

The following variables were used throughout the analysis:

ID – Unique donor identifier.

First_Name – Donor first name.

Last_Name – Donor last name.

Gender – Donor gender.

State – Geographic location.

Job_Field – Professional category.

Donation – Donation amount.

Donation_Frequency – Frequency of donor contributions.

University – Educational background.

Car – Vehicle preference.

Movie_Genre – Entertainment preference.

Second_Language – Additional language spoken.

Assumptions:

- Each donor ID is unique.
- Donation values are correctly recorded.
- Missing values represent unavailable information rather than data-entry errors.
- Records accurately represent historical donor behavior.

Data Quality Assessment

Data quality is a critical prerequisite for reliable analysis. Several validation procedures were performed.

Record Validation:

- Donation records: 1000
- Donor records: 1000
- Integrated records after join: 1000

Missing Data:

- University missing values: 459
- Second language missing values: 667

Quality Checks Performed:

- Verification of unique donor IDs.
- Validation of table joins.
- Identification of missing values.
- Review of donation value ranges.
- Detection of potential inconsistencies.

The overall quality of the dataset is suitable for business analytics. Missing values are limited primarily to descriptive demographic variables and do not significantly affect donation calculations.

Methodology

Query-Method

The project followed a structured analytics process:

1. Data Extraction
SQL scripts were used to retrieve records from the source tables.
2. Data Integration
Donation and donor datasets were joined using donor ID.
3. Data Cleaning
Missing values, duplicates, and inconsistencies were evaluated.
4. Descriptive Analysis
SQL aggregate functions were used to calculate totals, averages, counts, minimums, and maximums.
5. Segmentation Analysis
Donors were grouped by state, gender, job field, and donation frequency.
6. Business Interpretation
Statistical findings were translated into actionable fundraising recommendations.

This methodology ensures both technical accuracy and business relevance.

Visualization-Method

1. Imported the donor and donation dataset into Power BI Desktop.
2. Reviewed the dataset structure to understand donor demographics, donation amounts, job fields, gender, state, and donation frequency.
3. Performed data cleaning in Power Query by checking data types, handling missing values, and validating records.
4. Created calculated measures using DAX for key metrics such as Total Donations, Average Donation, Maximum Donation, Minimum Donation, and Total Donors.
5. Built an Executive Overview page to provide a high-level summary of donation performance and donor activity.
6. Developed a Job Field Analysis page to evaluate donation contributions and donor distribution across professions.
7. Created a Geographical Analysis page to identify the highest and lowest donating states and visualize regional donation patterns.

8. Designed a Gender Analysis page to compare donation behavior and contribution levels between male and female donors.
9. Built a Data Quality dashboard to assess missing values, measure data completeness, and identify areas requiring data improvement.
10. Applied interactive features and dashboard design principles, including page navigation, KPIs, filters, and consistent visual formatting to enhance user experience and decision-making.

Analysis & Findings

The integrated dataset provides a detailed view of donor behavior. Total donations equal \$249,085. The average donation amount is \$249.09. The minimum donation recorded is \$5, while the maximum donation is \$500.

Gender Analysis:

Donation totals were evaluated by gender to determine whether fundraising performance varies across demographic groups. The analysis indicates that both genders contribute significantly to overall fundraising revenue, suggesting that future campaigns should maintain broad demographic appeal.

Geographic Analysis:

State-level analysis revealed meaningful differences in donation volume. Geographic segmentation allows organizations to identify strong-performing regions and allocate marketing resources more effectively.

Job Field Analysis:

Donors from different professional backgrounds contribute at varying levels. Understanding occupational patterns can support more personalized donor engagement strategies.

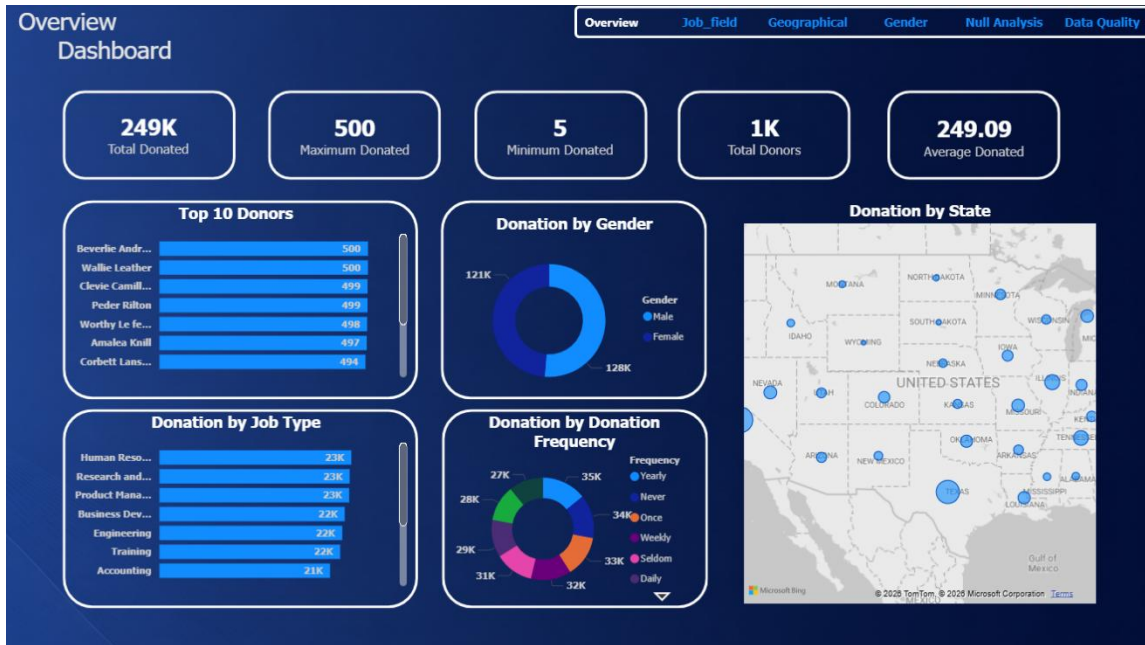
Donation Frequency Analysis:

Donation frequency is one of the strongest indicators of donor value. Recurring donors typically generate higher lifetime value compared with one-time contributors. This finding reinforces the importance of donor retention initiatives and recurring donation programs.

High-Value Donor Analysis:

The top donor segment contributes a disproportionately large share of total donations. Maintaining relationships with these donors should be considered a strategic priority.

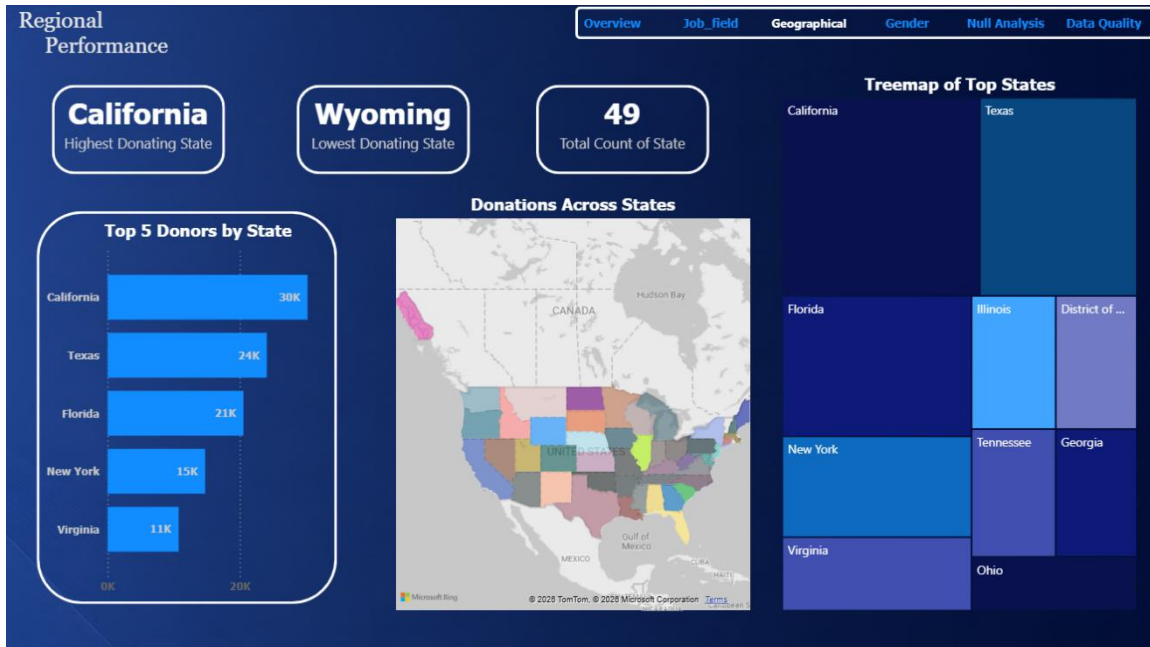
Overview of Analysis



Job Field Analysis



Geographical Analysis



Gender Analysis



Null Analysis



Key Insights

- Donation frequency is strongly associated with fundraising performance.
- Geographic differences suggest opportunities for localized fundraising campaigns.
- Professional background can be used as a donor segmentation variable.
- Data quality is sufficiently strong to support business decision-making.
- Retention of recurring donors is likely to generate higher long-term fundraising returns.

Recommendations

1. Increase investment in donor retention programs because recurring donors generate sustainable revenue streams.
2. Develop geographically targeted fundraising campaigns in high-performing states to maximize return on fundraising investment.
3. Implement donor segmentation strategies based on occupation, demographics, and donation behavior.
4. Improve data collection processes to reduce missing demographic information and enable more advanced analytics.
5. Establish executive dashboards and KPI monitoring processes to track donation trends and donor engagement over time.
6. Introduce personalized communication strategies tailored to donor preferences and behavioral characteristics.
7. The organization should capture a Donation Date field for every transaction.

Limitations & Assumptions

Several limitations should be considered. The dataset represents historical donor records and may not fully predict future donor behavior. Missing values reduce the ability to perform deeper demographic segmentation. External variables such as economic conditions, fundraising campaigns, seasonal effects, and communication activities were not available for analysis. Consequently, results should be interpreted within the context of available data.

Data Limitations

One significant limitation of this dataset is the absence of a date or timestamp field for donations. Because donation transactions are not associated with specific dates, it is not possible to:

- Analyze donation trends over time.
- Identify seasonal donation patterns.
- Determine which months or quarters generate the highest or lowest donations.
- Track donor behavior over specific periods.
- Measure donor retention and engagement trends.
- Evaluate the impact of fundraising campaigns over time.

Business Impact

Without timestamp information, organizations cannot effectively answer critical questions such as:

- When do donors contribute the most?
- Which quarter generates the highest donation revenue?
- Are donations increasing or declining over time?
- Which donors are becoming less active or more engaged?
- What periods require additional fundraising efforts?

Appendix

- A. SQL joins between Donation_Data and Donor_Data.
- B. Aggregation queries for donation totals.
- C. Data quality validation queries.
- D. Donor segmentation queries.
- E. Frequency analysis and ranking queries.
- F. Power Bi for Visualization and recommendation.

These SQL procedures formed the analytical foundation for all findings, Power Bi form the visualization of the findings from SQL and recommendations presented in this report.